

Debt Expansion Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Expansion Restraint (DER), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DER achieves an annualized gross (net) Sharpe ratio of 0.48 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (19) bps/month with a t-statistic of 3.22 (2.54), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets) is 22 bps/month with a t-statistic of 3.01.

1 Introduction

Production-based asset pricing models posit that firm investment decisions and production technologies fundamentally drive cross-sectional variation in expected returns. While substantial progress has been made in linking firm characteristics to risk premia through the lens of optimal investment theory, significant gaps remain in understanding how firms’ financing constraints and debt capacity decisions affect their exposure to systematic productivity shocks. The interaction between debt financing choices and investment frictions represents a critical yet underexplored dimension of production-based risk, particularly as firms navigate trade-offs between preserving financial flexibility and exploiting growth opportunities in the presence of capital market frictions.

In a production economy with investment adjustment costs, firms that exercise restraint in debt expansion signal lower marginal costs of capital and reduced exposure to negative productivity shocks. Following the q-theory framework of [Cochrane \(1991\)](#) and [Zhang \(2005\)](#), firms optimize investment by equating the marginal cost of investment to the marginal q, where financing frictions directly affect the shadow cost of capital. When firms limit debt expansion despite available borrowing capacity, they effectively maintain lower operating leverage and preserve flexibility to absorb future productivity shocks without costly external financing ([Hennessy and Whited, 2007](#)). This restraint in debt usage proxies for firms with higher productivity persistence and lower conditional risk premia, as these firms face lower costs of adjusting their capital stock in response to technology shocks ([Liu et al., 2009](#)).

The production-based interpretation of debt expansion restraint connects directly to the investment-based factor models through the channel of financial constraints on real investment. Firms exhibiting debt restraint maintain spare debt capacity that reduces the wedge between internal and external costs of capital, effectively lowering their exposure to aggregate investment shocks ([Gould and Jiao, 2023](#)). In

equilibrium, these firms command lower expected returns because their production technologies exhibit lower sensitivity to systematic variations in the stochastic discount factor that prices investment risk (Lin and Zhang, 2013). The cross-sectional variation in debt expansion restraint thus captures heterogeneity in firms’ conditional betas with respect to investment-specific technology shocks.

Furthermore, debt expansion restraint serves as a state variable for time-varying risk premia in production economies with financial frictions. During periods of aggregate productivity decline, firms with greater debt restraint possess superior ability to maintain optimal investment policies without incurring the deadweight costs of financial distress or forced deleveraging (Bhandari and Evans, 2023). This mechanism generates predictable variation in expected returns that aligns with the conditional CAPM in production economies, where the pricing kernel depends on the marginal utility of wealth for the representative firm facing investment adjustment costs (Balvers and Huang, 2009). The signal therefore captures both cross-sectional differences in unconditional risk loadings and time-series variation in the price of investment risk.

We find strong evidence that debt expansion restraint predicts cross-sectional stock returns. The value-weighted long-short DER strategy generates an average monthly return of 26 basis points with a t-statistic of 3.45, yielding an annualized Sharpe ratio of 0.48. The strategy’s performance remains robust across multiple factor model specifications, with the monthly alpha relative to the Fama-French six-factor model equaling 25 basis points with a t-statistic of 3.22. These results demonstrate both statistical and economic significance, as the strategy’s risk-adjusted returns persist after controlling for market, size, value, profitability, investment, and momentum factors.

The predictive power of debt expansion restraint extends across the size spectrum and survives stringent robustness tests. Among the largest quintile of stocks by

market capitalization, the DER strategy achieves an average return of 34 basis points per month with a t-statistic of 3.65, indicating that the anomaly is not confined to small, illiquid stocks. After accounting for transaction costs using the composite bid-ask spread measure, the net Sharpe ratio equals 0.37, placing the strategy in the 94th percentile of the anomaly distribution. The strategy maintains positive and significant net generalized alphas across all five standard factor models, confirming that it expands the mean-variance efficient frontier even after implementation costs.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DER strategy retains an alpha of 22 basis points per month with a t-statistic of 3.01. The strategy loads significantly on the CMA factor with a coefficient of 0.24 (t-statistic of 4.66), consistent with the production-based interpretation that debt restraint captures variation in investment frictions. When combined with 155 existing anomalies using the average rank method, adding DER increases the terminal value of a dollar invested from \$993.77 to \$1,243.77, demonstrating meaningful incremental value beyond the existing factor zoo.

This paper contributes to the production-based asset pricing literature by establishing debt expansion restraint as a robust predictor of cross-sectional returns that captures firms' heterogeneous exposure to investment frictions and productivity shocks. While [Zhang \(2005\)](#) and subsequent studies focus primarily on investment quantities and profitability measures, we demonstrate that financing restraint provides complementary information about firms' production technologies and conditional risk premia. Our findings extend the investment-based framework of [Liu et al. \(2009\)](#) by showing how debt capacity decisions reflect optimal dynamic investment policies under financial frictions. Unlike existing leverage-based predictors that emphasize financial distress risk, debt expansion restraint captures forward-looking managerial decisions about maintaining financial flexibility to accommodate future investment opportunities.

Methodologically, we employ the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#) to establish that debt expansion restraint satisfies stringent criteria for inclusion in the cross-section of expected return predictors. The anomaly’s gross Sharpe ratio of 0.48 exceeds 87% of documented anomalies, while its net Sharpe ratio of 0.37 surpasses 94% of strategies after transaction costs. These results address recent concerns about data mining and implementation costs raised by [Hou et al. \(2020\)](#) regarding the replicability crisis in empirical asset pricing. By demonstrating significant alphas relative to both standard factor models and closely related anomalies from the factor zoo, we establish that debt expansion restraint contains unique information about expected returns not captured by existing characteristics.

Our results have broader implications for understanding the joint dynamics of corporate financing and investment decisions in production economies. The robust performance of debt expansion restraint across size quintiles and market conditions suggests that financial flexibility represents a first-order consideration in firms’ production decisions, consistent with dynamic models of investment under uncertainty. The finding that debt restraint predicts returns even among large, liquid stocks indicates that production-based risks operate broadly across the cross-section rather than being confined to financially constrained firms. These insights inform both theoretical models of production-based asset pricing that incorporate financial frictions and practical applications in factor investing and risk management.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual obser-

vations to capture year-over-year changes in firms' long-term debt issuance activities. construction of our Debt Expansion Restraint signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, as reported in the firm's statement of cash flows. This variable captures new long-term borrowing activities and excludes refinancing of existing debt that does not result in net cash proceeds. Total assets (AT) represents the book value of all assets reported on the firm's balance sheet at fiscal year-end, providing a comprehensive measure of firm size that we use to scale the debt issuance changes. construct the Debt Expansion Restraint signal by calculating the year-over-year change in long-term debt issuance, scaled by lagged total assets: $(DLTIS(t) - DLTIS(t-1)) / AT(t-1)$. This measure captures the change in a firm's debt issuance activity relative to its size, with positive values indicating acceleration in debt issuance and negative values suggesting restraint in new debt financing. The scaling by lagged assets ensures comparability across firms of different sizes and mitigates the influence of firm growth on the measure. We compute this signal using fiscal year-end values and require non-missing values for long-term debt issuance in consecutive years and lagged assets to ensure data quality. Firms with zero or negative lagged assets are excluded from the analysis.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DER signal. Panel A plots the time-series of the mean, median, and interquartile range for DER. On average, the cross-sectional mean (median) DER is -0.03 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DER data. The signal's interquartile range spans -0.06 to 0.05. Panel B of Figure 1 plots the time-series of the coverage of the DER signal for the CRSP universe. On average, the DER signal

is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does DER predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DER using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DER portfolio and sells the low DER portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DER strategy earns an average return of 0.26% per month with a t-statistic of 3.45. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.25% to 0.31% per month and have t-statistics exceeding 3.22 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 4.66 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 535 stocks and an average market capitalization of at least \$1,410 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 21 bps/month with a t-statistics of 4.67. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-20bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.82. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DER trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DER strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DER, as well as average returns and alphas for long/short trading DER strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DER strategy achieves an average return of 34 bps/month with a t-statistic of 3.65. Among these large cap stocks, the alphas for the DER strategy relative to the five most common factor models range from 25 to 36 bps/month with t-statistics between 2.64 and 3.90.

5 How does DER perform relative to the zoo?

Figure 2 puts the performance of DER in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DER strategy falls in the distribution. The DER strategy’s gross (net) Sharpe ratio of 0.48 (0.37) is greater than 87% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DER strategy (red line).² Ignoring trading costs, a \$1 invested in the DER strategy would have yielded \$3.21 which ranks the DER strategy in the top 7% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the DER strategy would have yielded \$1.92 which ranks the DER strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DER relative to those. Panel A shows that the DER strategy gross alphas fall between the 58 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DER strategy has a positive net generalized alpha for five out of the five factor models. In these cases DER ranks between the 76 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DER add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DER with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DER or at least to weaken the power DER has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DER conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DER} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DER,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DER. Stocks are finally grouped into five DER portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DER trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DER and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DER signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on DER is -1.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the DER strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DER strategy earns alphas that range from 22-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DER trading strategy achieves an alpha of 22bps/month with a t-statistic of 3.01.

7 Does DER add relative to the whole zoo?

Finally, we can ask how much adding DER to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DER signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DER grows to \$1243.77.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DER is available.

8 Conclusion

This study provides robust evidence that Debt Expansion Restraint (DER) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DER-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DER delivers an annualized gross Sharpe ratio of 0.48, which remains economically significant at 0.37 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 25 basis points gross (19 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds, underscores its statistical and economic importance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of twelve factors, including six closely related financing and investment strategies. The persistence of a 22 basis points monthly alpha with a t-statistic of 3.01 in this demanding specification suggests that DER captures unique information about future returns not subsumed by existing factors.

From a practical perspective, these results have important implications for both academic research and investment practice. The economic magnitude of the returns, combined with their statistical significance even after transaction costs, suggests that DER could be valuable for portfolio construction and risk management. The signal appears to identify firms whose debt management policies contain predictive information about future performance, potentially reflecting either managerial discipline, financial flexibility, or market mispricing of leverage decisions.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and

capacity constraints in real-world trading environments may further erode returns. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different economic regimes or market structures. Third, the underlying economic mechanism driving the DER anomaly remains partially unexplained, leaving open questions about whether the returns represent compensation for systematic risk or market inefficiency.

Future research should explore several promising avenues. Investigation into the economic channels through which debt expansion restraint predicts returns would enhance our theoretical understanding of this anomaly. Cross-country and out-of-sample validation would establish the signal’s robustness across different institutional settings and time periods. Additionally, examining the interaction between DER and corporate governance, macroeconomic conditions, or industry characteristics could reveal important conditional relationships. Finally, exploring the optimal combination of DER with other signals and investigating its performance in different asset classes would provide valuable insights for practical implementation. As markets evolve and new data becomes available, continued evaluation of DER’s predictive power will be essential for understanding its long-term viability as an investment signal.

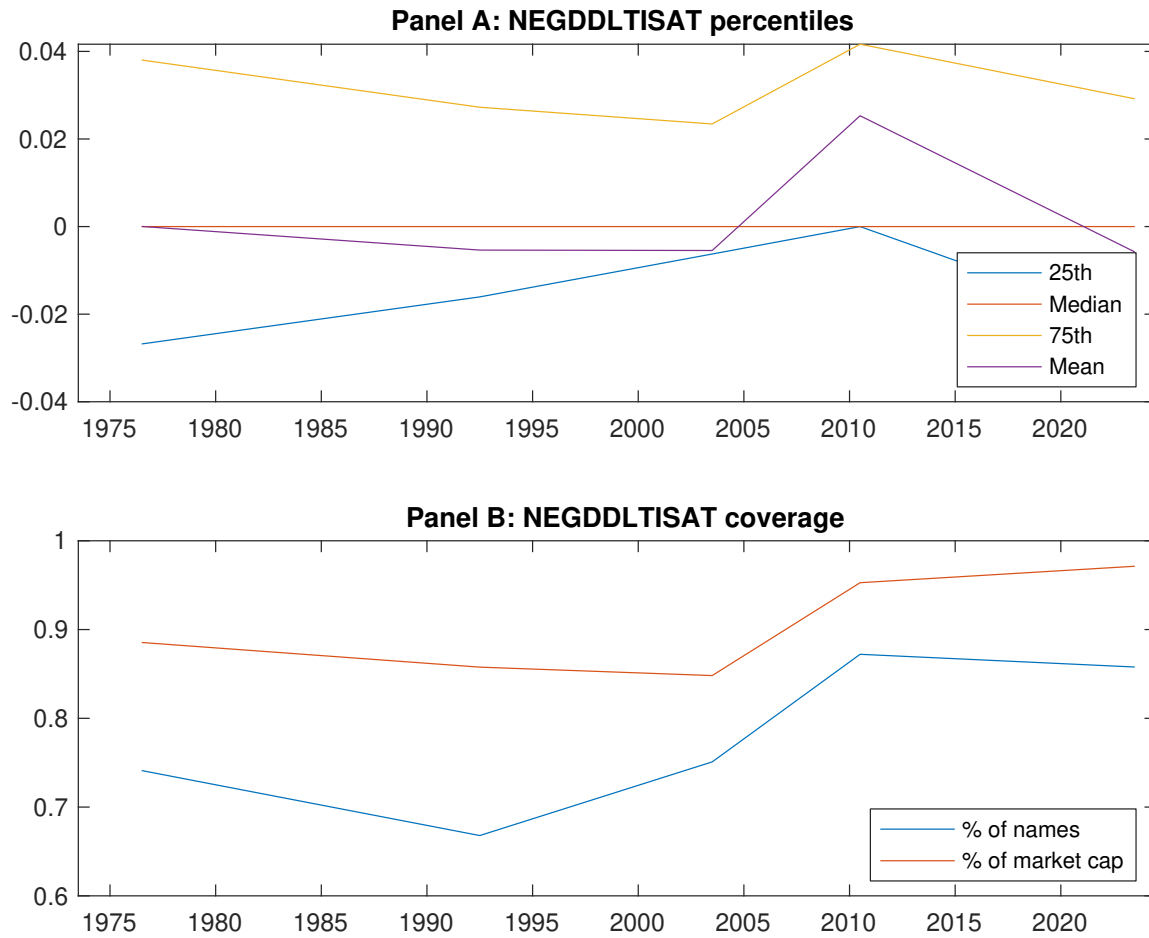


Figure 1: Times series of DER percentiles and coverage. This figure plots descriptive statistics for DER. Panel A shows cross-sectional percentiles of DER over the sample. Panel B plots the monthly coverage of DER relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DER. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DER-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.59 [2.81]	0.64 [3.53]	0.62 [3.13]	0.72 [4.04]	0.85 [4.26]	0.26 [3.45]
α_{CAPM}	-0.12 [-2.23]	0.03 [0.56]	-0.05 [-0.89]	0.11 [2.49]	0.18 [3.11]	0.30 [4.02]
α_{FF3}	-0.12 [-2.22]	-0.02 [-0.40]	-0.00 [-0.02]	0.09 [2.02]	0.19 [3.49]	0.31 [4.10]
α_{FF4}	-0.10 [-1.92]	0.01 [0.12]	0.04 [0.82]	0.06 [1.31]	0.17 [3.05]	0.27 [3.58]
α_{FF5}	-0.12 [-2.25]	-0.06 [-1.32]	0.05 [0.92]	0.01 [0.15]	0.15 [2.65]	0.27 [3.53]
α_{FF6}	-0.11 [-2.05]	-0.04 [-0.87]	0.08 [1.51]	-0.01 [-0.26]	0.13 [2.41]	0.25 [3.22]
Panel B: Fama and French (2018) 6-factor model loadings for DER-sorted portfolios						
β_{MKT}	1.06 [83.62]	0.98 [96.26]	0.99 [77.85]	0.98 [97.21]	1.02 [79.29]	-0.04 [-2.21]
β_{SMB}	0.16 [8.39]	-0.14 [-9.08]	-0.02 [-0.87]	-0.04 [-2.72]	0.14 [7.51]	-0.01 [-0.55]
β_{HML}	-0.03 [-1.21]	0.16 [8.06]	-0.10 [-4.23]	0.02 [1.22]	-0.14 [-5.53]	-0.11 [-3.16]
β_{RMW}	0.08 [3.16]	0.12 [5.75]	-0.05 [-2.01]	0.11 [5.53]	0.02 [0.82]	-0.06 [-1.66]
β_{CMA}	-0.09 [-2.35]	0.00 [0.02]	-0.11 [-2.90]	0.16 [5.38]	0.15 [4.09]	0.24 [4.66]
β_{UMD}	-0.02 [-1.24]	-0.03 [-3.09]	-0.05 [-4.05]	0.03 [2.78]	0.02 [1.46]	0.03 [1.95]
Panel C: Average number of firms (n) and market capitalization (me)						
n	671	535	1060	590	641	
me (\$10 ⁶)	1410	3195	2218	3114	1416	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DER strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [3.45]	0.30 [4.02]	0.31 [4.10]	0.27 [3.58]	0.27 [3.53]	0.25 [3.22]
Quintile	NYSE	EW	0.21 [4.67]	0.24 [5.27]	0.22 [4.91]	0.20 [4.46]	0.21 [4.59]	0.20 [4.32]
Quintile	Name	VW	0.25 [3.42]	0.29 [4.11]	0.30 [4.23]	0.26 [3.59]	0.25 [3.42]	0.22 [3.04]
Quintile	Cap	VW	0.21 [3.29]	0.25 [3.82]	0.25 [3.78]	0.20 [3.10]	0.17 [2.70]	0.15 [2.29]
Decile	NYSE	VW	0.23 [2.42]	0.31 [3.30]	0.28 [2.94]	0.24 [2.49]	0.12 [1.33]	0.11 [1.19]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.20 [2.64]	0.23 [3.13]	0.24 [3.17]	0.22 [2.91]	0.21 [2.76]	0.19 [2.54]
Quintile	NYSE	EW	-0.05 [-0.82]					
Quintile	Name	VW	0.19 [2.58]	0.24 [3.31]	0.24 [3.38]	0.22 [3.06]	0.20 [2.80]	0.18 [2.55]
Quintile	Cap	VW	0.16 [2.48]	0.20 [3.06]	0.19 [2.98]	0.17 [2.63]	0.13 [2.09]	0.12 [1.81]
Decile	NYSE	VW	0.16 [1.66]	0.24 [2.50]	0.21 [2.19]	0.19 [1.97]	0.08 [0.90]	0.07 [0.75]

Table 3: Conditional sort on size and DER

This table presents results for conditional double sorts on size and DER. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DER. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DER and short stocks with low DER. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DER Quintiles					DER Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.65 [2.36]	0.86 [3.20]	0.94 [3.41]	0.87 [3.18]	0.83 [2.91]	0.17 [1.86]	0.20 [2.16]	0.19 [1.99]	0.14 [1.44]	0.15 [1.62]	0.12 [1.28]
	(2)	0.72 [2.69]	0.90 [3.65]	0.82 [3.22]	0.88 [3.58]	0.87 [3.42]	0.14 [1.78]	0.18 [2.26]	0.15 [1.96]	0.16 [1.98]	0.14 [1.73]	0.14 [1.78]
	(3)	0.78 [3.06]	0.79 [3.62]	0.81 [3.35]	0.85 [3.79]	0.90 [3.83]	0.12 [1.52]	0.19 [2.41]	0.18 [2.28]	0.14 [1.75]	0.16 [1.98]	0.13 [1.62]
	(4)	0.73 [3.18]	0.78 [3.75]	0.84 [3.84]	0.74 [3.55]	0.91 [4.15]	0.17 [2.31]	0.21 [2.75]	0.20 [2.68]	0.18 [2.28]	0.19 [2.39]	0.17 [2.12]
	(5)	0.50 [2.56]	0.60 [3.36]	0.58 [2.94]	0.61 [3.35]	0.83 [4.42]	0.34 [3.65]	0.36 [3.88]	0.36 [3.90]	0.29 [3.15]	0.29 [3.13]	0.25 [2.64]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DER Quintiles					DER Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	392	393	393	393	389	37	34	32	33	35	
	(2)	107	107	107	108	107	61	61	60	61	61	
	(3)	76	77	76	76	77	107	110	105	107	108	
	(4)	64	65	64	65	64	229	240	229	235	230	
(5)	59	59	59	59	59	1453	2220	1800	2187	1517		

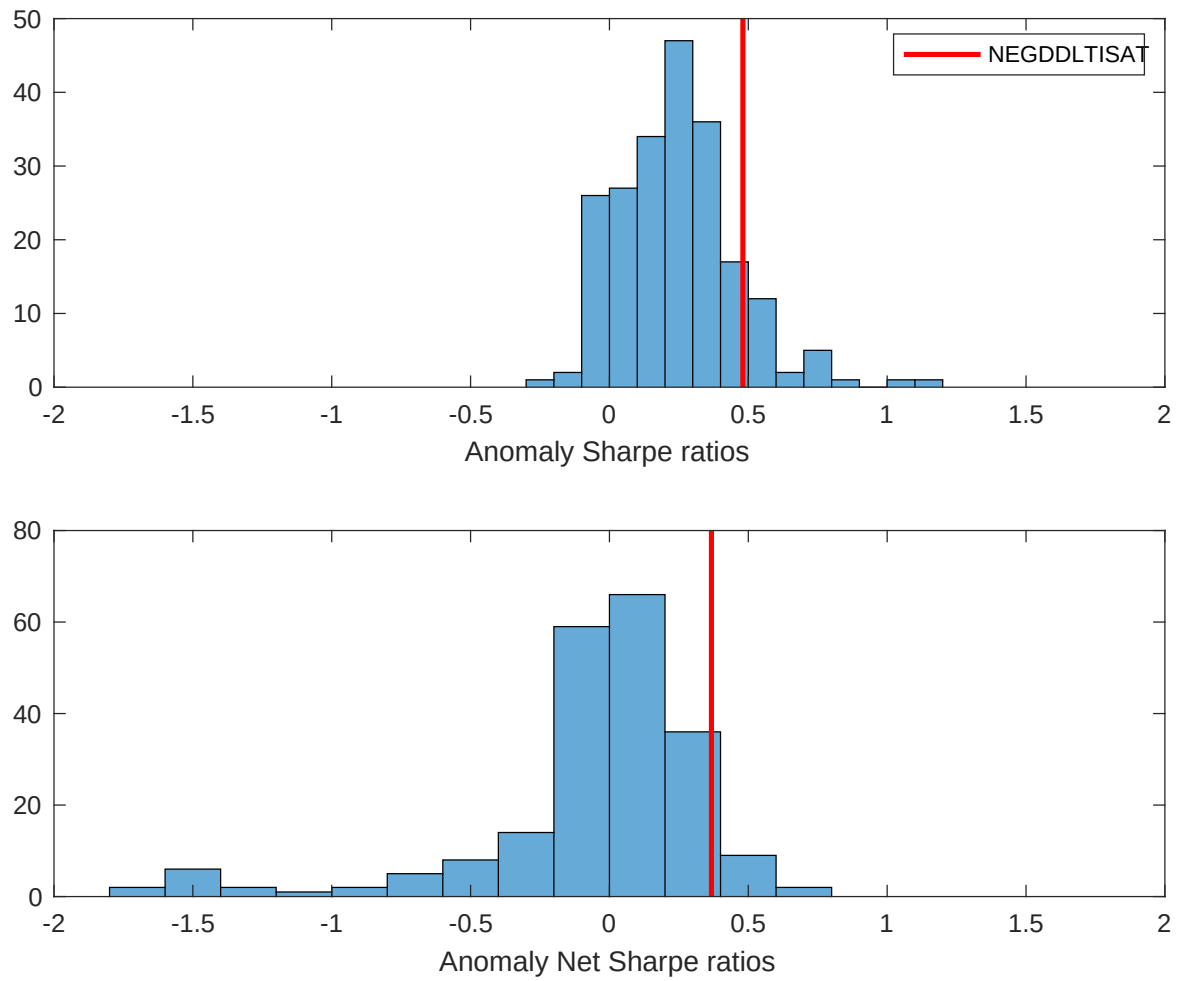


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DER with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

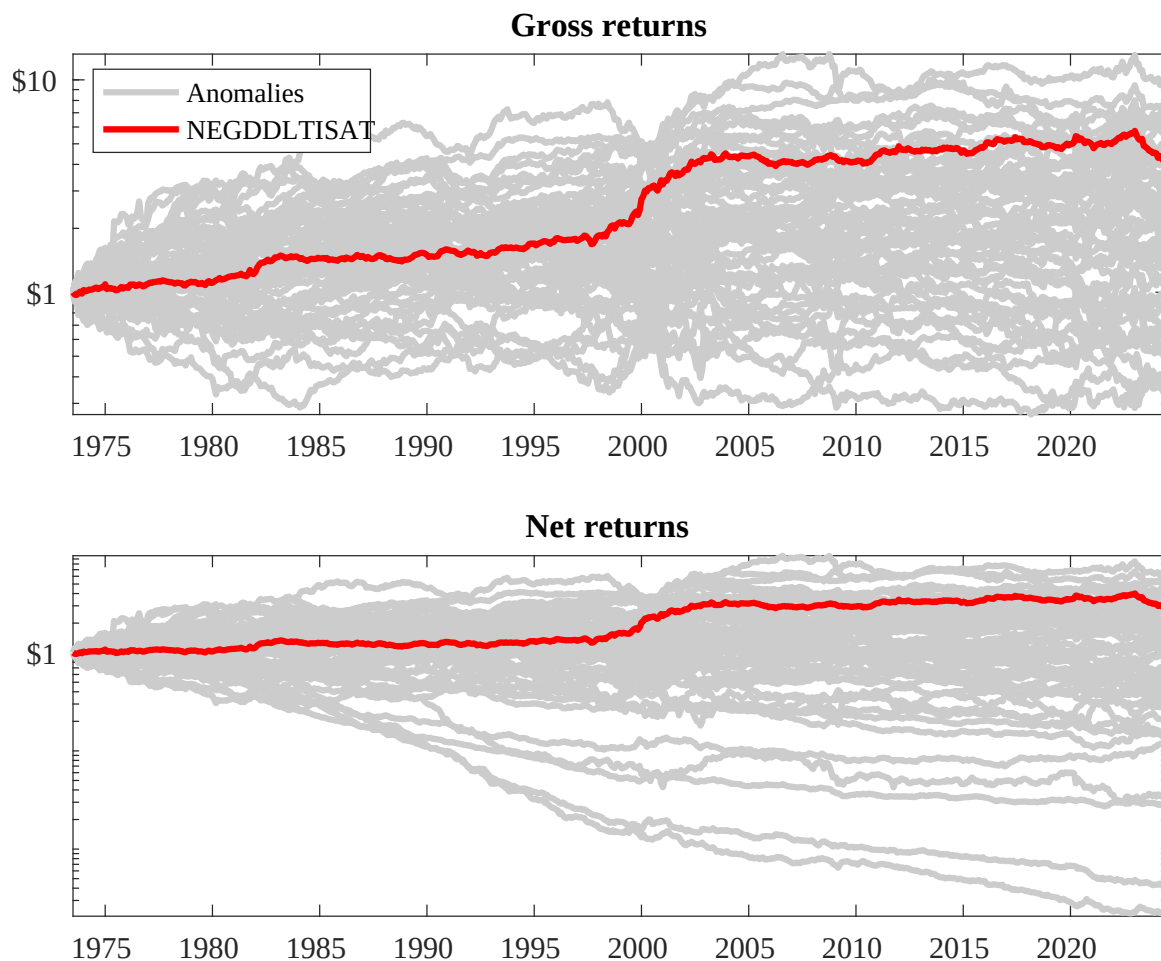
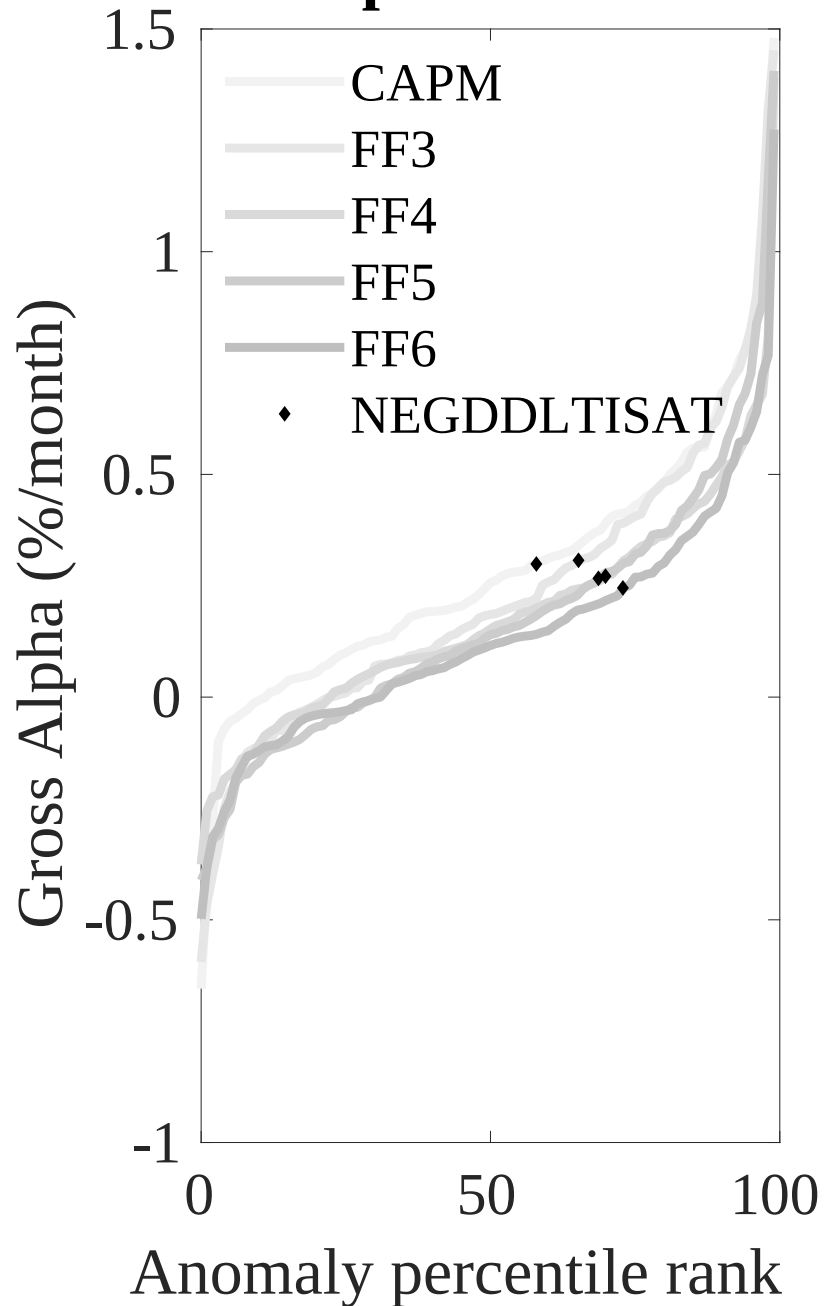


Figure 3: Dollar invested.

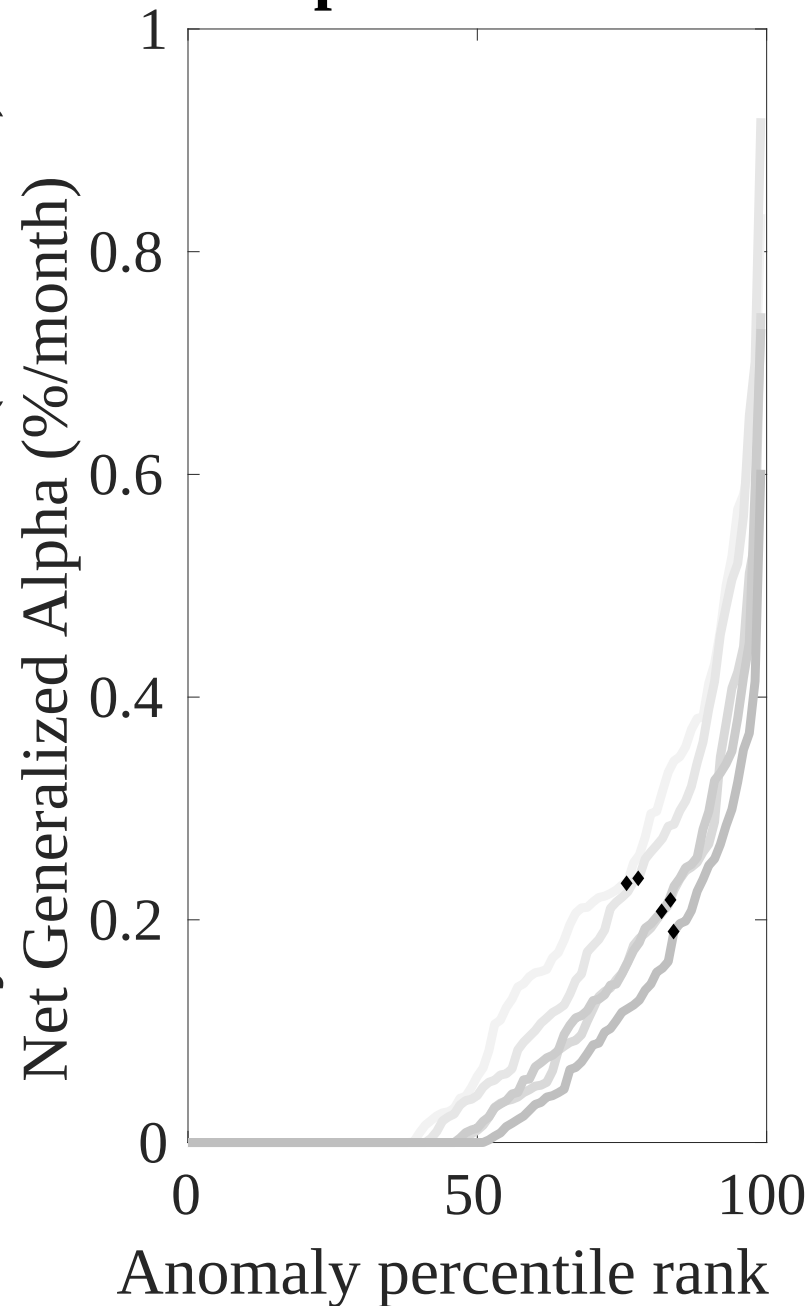
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DER trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



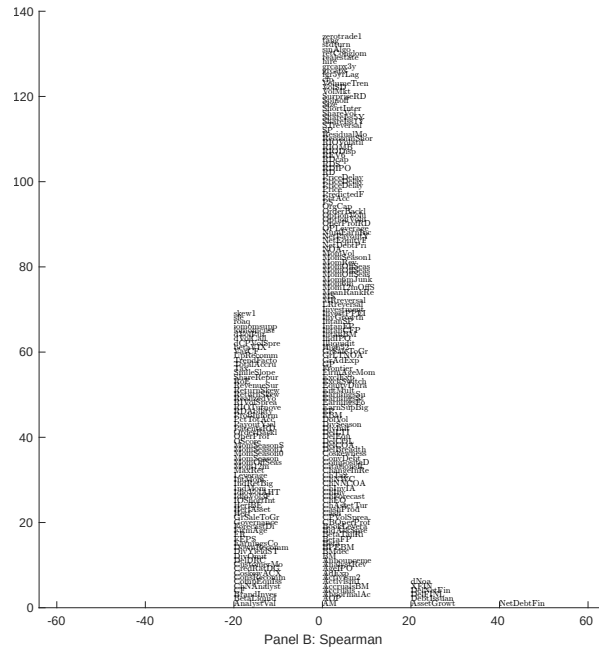
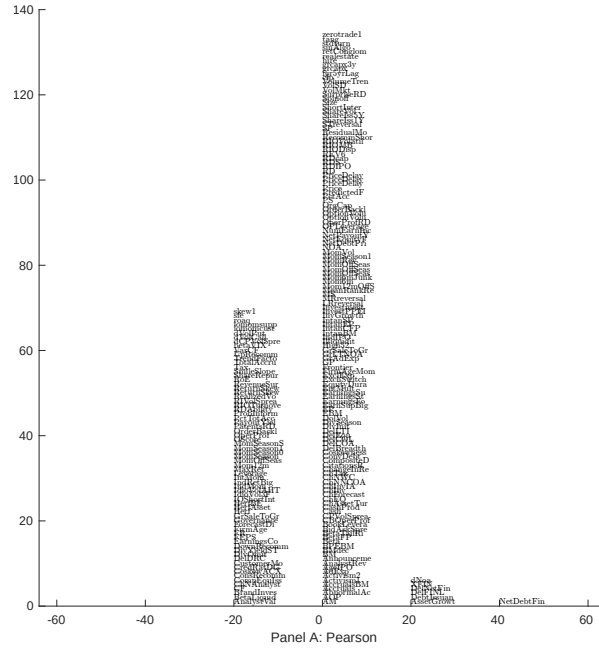


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DER. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

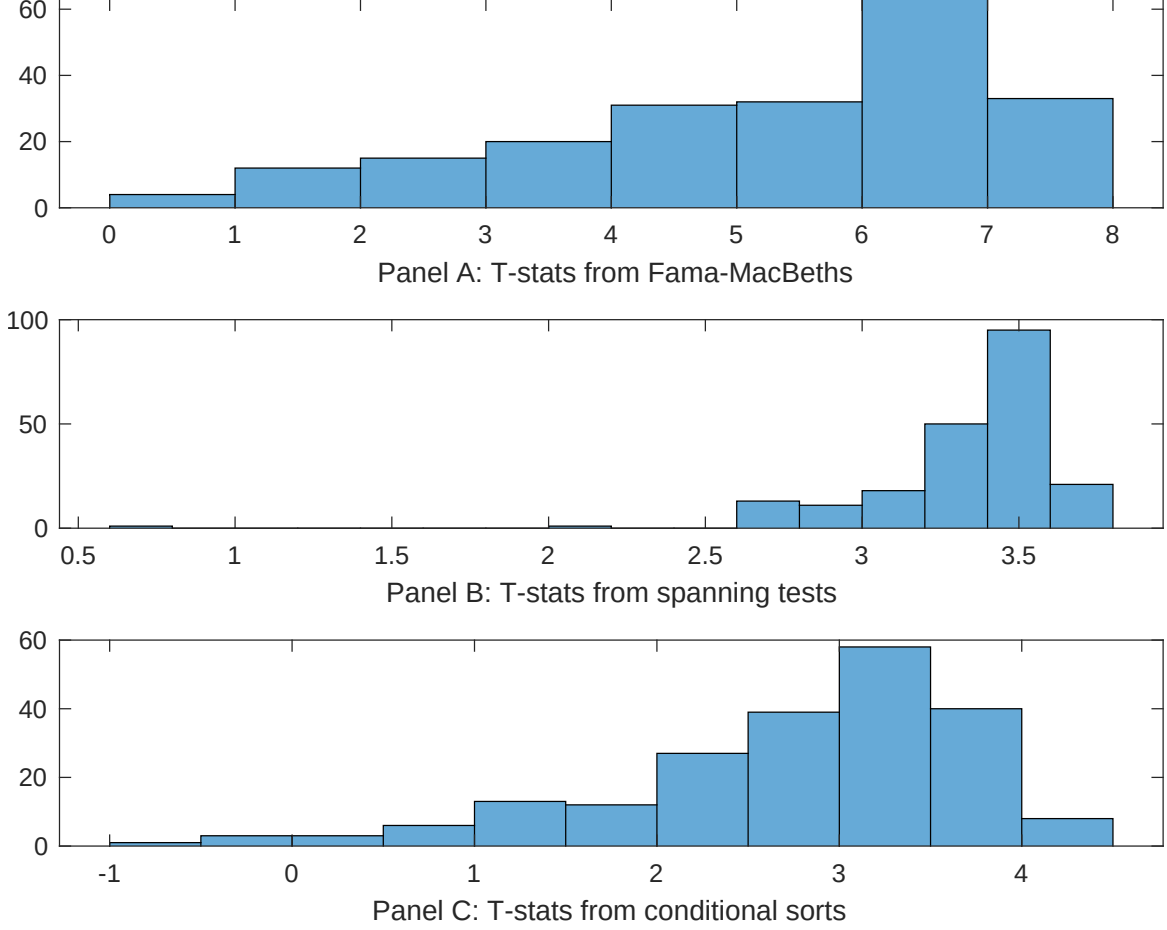


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DER conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DER} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DER,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DER. Stocks are finally grouped into five DER portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DER trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DER. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.33]	0.13 [5.36]	0.14 [5.32]	0.14 [5.69]	0.14 [5.73]	0.14 [5.65]	0.15 [5.75]
DER	0.12 [1.21]	0.10 [1.11]	0.51 [5.11]	0.20 [1.88]	0.21 [2.33]	0.29 [0.31]	-0.14 [-1.22]
Anomaly 1	0.20 [8.54]						0.11 [1.59]
Anomaly 2		0.18 [9.14]					-0.10 [-2.20]
Anomaly 3			0.40 [6.87]				0.21 [0.35]
Anomaly 4				0.19 [6.09]			0.11 [2.16]
Anomaly 5					0.17 [7.67]		0.69 [2.29]
Anomaly 6						0.14 [9.71]	0.96 [4.63]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DER trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DER} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.22 [2.94]	0.22 [2.88]	0.24 [3.20]	0.22 [2.89]	0.23 [3.12]	0.23 [2.99]	0.22 [3.01]
Anomaly 1	19.46 [4.77]						10.38 [1.87]
Anomaly 2		16.73 [3.94]					10.42 [1.79]
Anomaly 3			10.13 [3.46]				9.19 [2.99]
Anomaly 4				13.69 [3.64]			9.30 [2.36]
Anomaly 5					9.28 [2.72]		4.22 [1.12]
Anomaly 6						4.34 [1.00]	-8.15 [-1.68]
mkt	-3.52 [-2.05]	-3.17 [-1.84]	-3.74 [-2.16]	-1.68 [-0.93]	-3.68 [-2.12]	-3.54 [-2.03]	-2.26 [-1.27]
smb	-2.96 [-1.13]	-2.96 [-1.12]	-0.13 [-0.05]	3.16 [1.10]	-1.21 [-0.46]	-1.16 [-0.44]	0.63 [0.22]
hml	-9.73 [-2.94]	-9.41 [-2.82]	-10.85 [-3.25]	-9.00 [-2.68]	-11.52 [-3.42]	-10.88 [-3.21]	-8.46 [-2.52]
rmw	-7.67 [-2.26]	-7.04 [-2.07]	-4.06 [-1.19]	-13.71 [-3.37]	-5.36 [-1.57]	-5.46 [-1.60]	-11.89 [-2.93]
cma	19.28 [3.76]	18.73 [3.57]	15.33 [2.69]	15.40 [2.74]	16.98 [2.95]	21.12 [3.51]	6.63 [1.02]
umd	2.34 [1.36]	2.14 [1.22]	2.55 [1.46]	3.50 [2.04]	3.57 [2.07]	3.45 [1.98]	1.26 [0.72]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	10	9	9	9	8	7	13

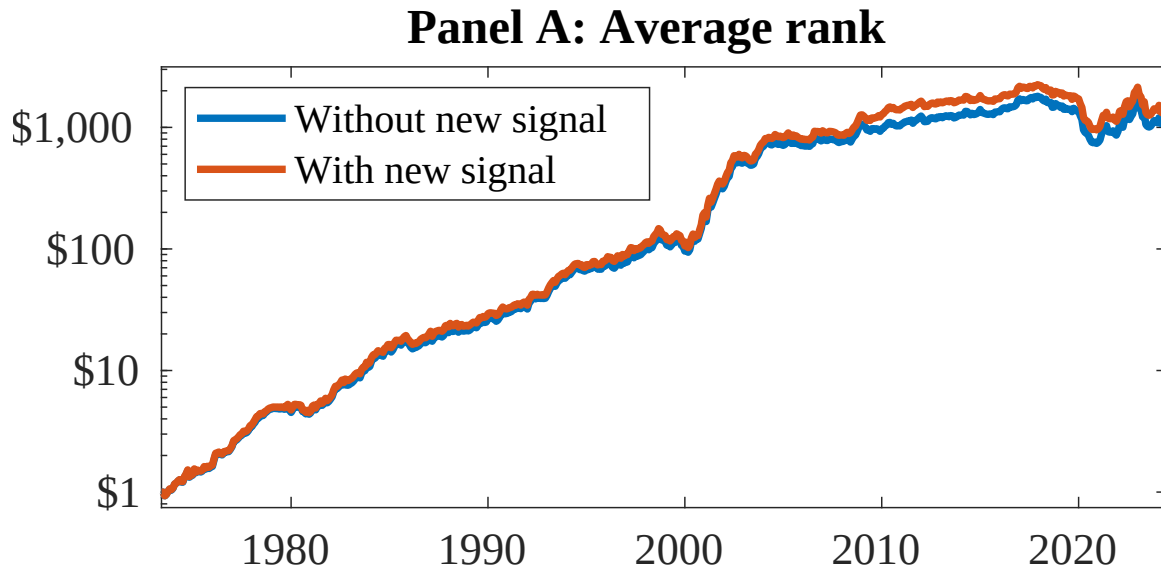


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DER. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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